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Supplementary material

Pointer usage on the Penn Treebank

For a qualitative analysis, we visualize how the pointer component is used within the pointer sentinel mixture model. The gate refers to the result of the gating function, with 1 indicating the RNN component is exclusively used whilst 0 indicates the pointer component is exclusively used. We begin with predictions that are using the RNN component primarily and move to ones that use the pointer component primarily.

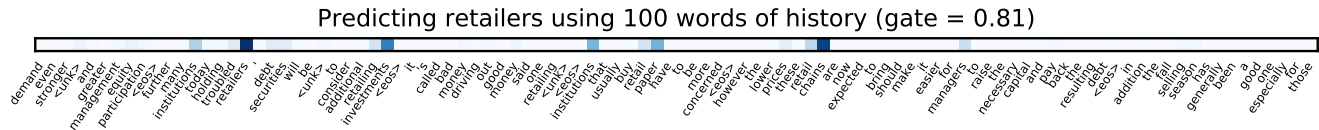


Figure 5. In predicting *the fall season has been a good one especially for those **retailers***, the pointer component suggests many words from the historical window that would fit - *retailers*, *investments*, *chains*, and *institutions*. The gate is still primarily weighted towards the RNN component however.

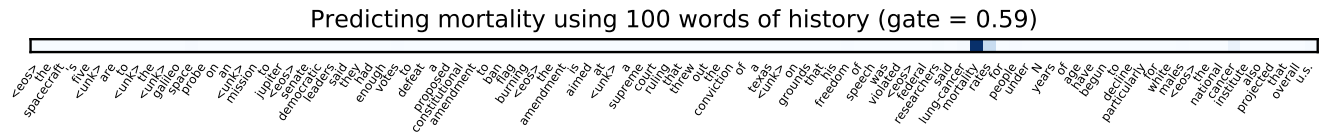


Figure 6. In predicting *the national cancer institute also projected that overall u.s. **mortality***, the pointer component is focused on *mortality* and *rates*, both of which would fit. The gate is still primarily weighted towards the RNN component.

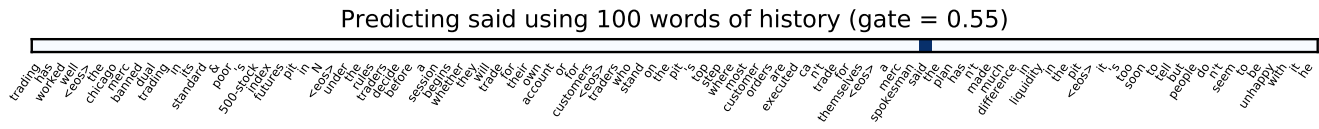


Figure 7. In predicting *people do n't seem to be unhappy with it he **said***, the pointer component correctly selects *said* and is almost equally weighted with the RNN component. This is surprising given how frequent the word **said** is used within the Penn Treebank.

Predicting billion using 100 words of history (gate = 0.44)

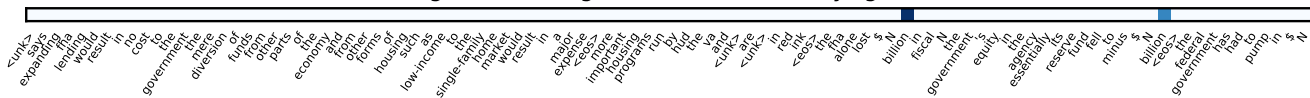


Figure 8. For predicting *the federal government has had to pump in \$ N billion*, the pointer component focuses on the recent usage of billion with highly similar context. The pointer component is also relied upon more heavily than the RNN component - surprising given the frequency of **billion** within the Penn Treebank and that the usage was quite recent.

Predicting noriega using 100 words of history (gate = 0.12)

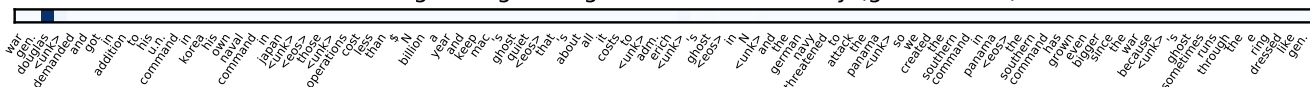


Figure 9. For predicting *<unk> 's ghost sometimes runs through the e ring dressed like gen. noriega*, the pointer component reaches 97 timesteps back to retrieve gen. douglas. Unfortunately this prediction is incorrect but without additional context a human would have guessed the same word. This additionally illustrates why the gating function must be integrated into the pointer component. The named entity *gen. douglas* would have fallen out of the window in only four more timesteps, a fact that the RNN hidden state would not be able to accurately retain for almost 100 timesteps.

Predicting iverson using 100 words of history (gate = 0.03)



Figure 10. For predicting *mr. iverson*, the pointer component has learned the ability to point to the last name of the most recent named entity. The named entity also occurs 45 timesteps ago, which is longer than the 35 steps that most language models truncate their backpropagation to.

Predicting rosenthal using 100 words of history (gate = 0.00)

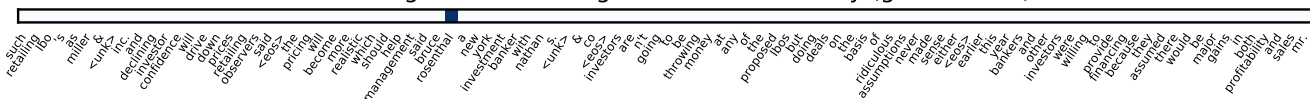


Figure 11. For predicting *mr. rosenthal*, the pointer is almost exclusively used and reaches back 65 timesteps to identify *bruce rosenthal* as the person speaking, correctly only selecting the last name.

Predicting integrated using 100 words of history (gate = 0.00)

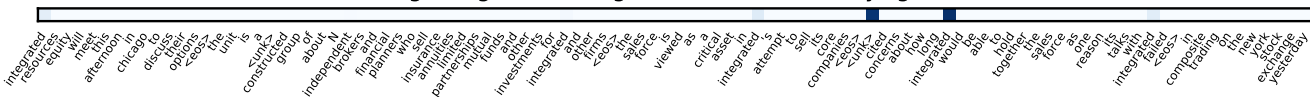


Figure 12. For predicting *in composite trading on the new york stock exchange yesterday integrated*, the company Integrated and the *<unk>* token are primarily attended to by the pointer component, with nearly the full prediction being determined by the pointer component.